M A T H E M A T I C S

MATH 00s

MATH 52 LEC 0.00

Course ID: 010375

Pre-University Calculus

The concepts included are limits, derivatives, antiderivatives and definite integrals. These concepts will be applied to solve problems of rates of change, maximum and minimum, curve sketching and areas. The classes of functions used to develop these concepts and applications are: polynomial, rational, trigonometric, exponential, and logarithmic.

Only offered Online

MATH 97 LEC 2.50

Course ID: 010113

Study Abroad

For studies at other universities under approved exchange agreements.

Department Consent Required

MATH 100s

MATH 103 LEC,TUT 0.50

Course ID: 006847

Introductory Algebra for Arts and Social Science

An introduction to applications of algebra to business, the behavioural sciences, and the social sciences. Topics will be chosen from linear equations, systems of linear equations, linear inequalities, functions, set theory, permutations and combinations, binomial theorem, probability theory. [Offered: F,W]

Prereq: Open only to students in the following faculties: ARTS, AHS or ENV. Not open to Acc'ting & Fin Mgt students.

Antireq: MATH 106, 114, 115, 136, 146, NE 112

Also offered Online

MATH 104 LEC,TUT 0.50

Course ID: 006848

Introductory Calculus for Arts and Social Science

An introduction to applications of calculus in business, the behavioural sciences, and the social sciences. The models studied will involve polynomial, rational, exponential, and logarithmic functions. The major concepts introduced to solve problems are rate of change, optimization, growth and decay, and integration. [Offered: F,W]

Prereq: Open only to students in the following Faculties: ARTS, AHS, ENV, SCI.

Antireq: MATH 127, 137, 147

Also offered Online

MATH 106 LEC,TUT 0.50

Course ID: 006869

Applied Linear Algebra 1

Systems of linear equations. Matrix algebra. Determinants. Introduction to vector spaces. Applications. [Offered: F,W,S]

Prereq: MATH 103 or 4U Calculus and Vectors; Not open to Computer Science students.

Antireq: MATH 114, 115, 136, 146, NE 112

Also offered Online

MATH 114 LEC,TUT 0.50

Course ID: 011645

Linear Algebra for Science

Vectors in 2- and 3-space and their geometry. Linear equations, matrices, and determinants. Introduction to vector spaces. Eigenvalues and diagonalization. Applications. Complex numbers. [Offered: F]

Prereq: 4U Calculus and Vectors; Science or Geomatics students only.

Antireq: MATH 106, 115, 136, 146, NE 112

Also offered Online

MATH 115 LEC,TST,TUT 0.50

Course ID: 006862

Linear Algebra for Engineering

Linear equations, matrices and determinants. Introduction to vector spaces. Eigenvalues and diagonalization. Applications. Complex numbers. [Offered: F]

Prereq: 4U Calculus and Vectors or 4U Mathematics of Data Management; Engineering students only.

Antireq: MATH 106, 114, 136, 146, NE 112

MATH 116 LEC,TST,TUT 0.50

Course ID: 006865

Calculus 1 for Engineering

Functions: review of polynomials, exponential, logarithmic, trigonometric. Operations on functions, curve sketching. Trigonometric identities, inverse functions. Derivatives, rules of differentiation. Mean Value Theorem, Newton's Method. Indeterminate forms and L'Hopital's rule, applications. Integrals, approximations, Riemann definite integral, Fundamental Theorems. Applications of the integral. [Offered: F]

Prereq: 4U Calculus and Vectors; Open to students in Engineering excluding Electrical and Computer Eng, Nanotechnology Eng, Software Eng and Systems Design Eng.

Antireq: MATH 117, 124, 127, 137, 147

MATH 117 LEC,TST,TUT 0.50

Course ID: 006866

Calculus 1 for Engineering

Functions of engineering importance; review of polynomial, exponential, and logarithmic functions; trigonometric functions and identities. Inverse functions (logarithmic and trigonometric). Limits and continuity. Derivatives, rules of differentiation; derivatives of elementary functions. Applications of the derivative, max-min problems, Mean Value Theorem. Antiderivatives, the Riemann definite integral, Fundamental Theorems. Methods of integration, approximation, applications, improper integrals. [Offered: F]

Prereq: 4U Calculus and Vectors; Open only to students in Electrical and Computer Engineering or Software Engineering or Nanotechnology Engineering.

Antireq: MATH 116, 124, 127, 137, 147

MATH 118 LEC,TST,TUT 0.50

Course ID: 006867

Calculus 2 for Engineering

Methods of integration: by parts, trigonometric substitutions, partial fractions; engineering applications, approximation of integrals, improper integrals. Linear and separable first order differential equations, applications. Parametric curves and polar coordinates, arc length and area. Infinite sequences and series, convergence tests, power series and applications. Taylor polynomials and series, Taylor's Remainder Theorem, applications. [Offered: W,S]

Prereq: One of MATH 116, 117, 127, 137, 147; Open only to students in Engineering excluding students in Electrical and Computer Eng, Nanotechnology Eng, Software Eng and Systems Design Eng.

Antireq: MATH 119, 128, 138, 148

MATH 119 LEC,TST,TUT 0.50

Course ID: 006868

Calculus 2 for Engineering

Elementary approximation methods: interpolation; Taylor polynomials and remainder; Newton's method, Landau order symbol, applications. Infinite series: Taylor series and Taylor's Remainder Theorem, geometric series, convergence test, power series, applications. Functions of several variables: partial derivatives, linear approximation and differential, gradient and directional derivative, optimization and Lagrange multipliers. Vector-valued functions: parametric representation of curves, tangent and normal vectors, line integrals and applications. [Offered: W,S]

Prereq: One of MATH 116, 117, 127, 137, 147; Open only to students in Electrical and Computer Engineering or Software Engineering or Nanotechnology Engineering.

Antireq: MATH 118, 128, 138, 148

MATH 124 LEC,TUT 0.50

Course ID: 012879

Calculus and Vector Algebra for Kinesiology

Review of trigonometry and basic algebra. Introduction to vectors in 2- and 3-space: sums, addition, dot products, cross products and angles between vectors. Solving linear systems in two and three variables. Functions of a real variable: powers, rational functions, trigonometric, exponential and logarithmic functions, their properties. Intuitive discussion of limits and continuity. Derivatives of elementary functions, derivative rules; applications to curve sketching, optimization. Relationships between distance, velocity, and acceleration. The definite integral, antiderivatives, the Fundamental Theorem of Calculus; change of variable and integration by parts; applications to area, centre of mass. [Offered: F]

Prereq: 4U Advanced Functions; Kinesiology students only.

Antireq: MATH 109, 116, 117, 127, 137, 147

Also offered Online

MATH 127 LEC,TST,TUT 0.50

Course ID: 006871

Calculus 1 for the Sciences

Functions of a real variable: powers, rational functions, trigonometric, exponential and logarithmic functions, their properties and inverses. Intuitive discussion of limits and continuity. Definition and interpretation of the derivative, derivatives of elementary functions, derivative rules and applications. Riemann sums and other approximations to the definite integral. Fundamental theorems and antiderivatives; change of variable. Applications to area, rates, average value. [Offered: F,W,S; online: F,W,S]

Prereq: MATH 104 or 4U Calculus and Vectors.

Antireq: MATH 109, 116, 117, 124, 137, 147

Also offered Online

MATH 128 LEC,TST,TUT 0.50

Course ID: 006872

Calculus 2 for the Sciences

Transforming and evaluating integrals; application to volumes and arc length; improper integrals. Separable and linear first order differential equations and applications. Introduction to sequences. Convergence of series; Taylor polynomials, Taylor's Remainder theorem, Taylor series and applications. Parametric/vector representation of curves; particle motion and arc length. Polar coordinates in the plane. [Offered: F,W,S; online: F,W,S]

Prereq: One of MATH 116, 117, 127, 137, 147.

Antireq: MATH 118, 119, 138, 148

Also offered Online

MATH 135 LEC,TST,TUT 0.50

Course ID: 006878

Algebra for Honours Mathematics

An introduction to the language of mathematics and proof techniques through a study of the basic algebraic systems of mathematics: the integers, the integers modulo n, the rational numbers, the real numbers, the complex numbers and polynomials. [Offered: F,W,S]

Prereq: 4U Calculus and Vectors or 4U Mathematics of Data Management; Honours Mathematics or Mathematics/BASE or Software Engineering students only.

Antireq: MATH 145

Also offered Online

MATH 136 LAB,LEC,TST,TUT 0.50

Course ID: 006879

Linear Algebra 1 for Honours Mathematics

Systems of linear equations, matrix algebra, elementary matrices, computational issues. Real n-space, vector spaces and subspaces, basis and dimension, rank of a matrix, linear transformations, and matrix representations. Determinants, eigenvalues and diagonalization, applications. [Offered: F,W,S; online: F,W,S]

Prereq: (MATH 135 with a grade of at least 60% or MATH 145; Honours Mathematics or Mathematics/BASE students) or Science Mathematical Physics students.

Antireq: MATH 106, 114, 115, 146, NE 112

Also offered Online

MATH 137 LAB,LEC,TST,TUT 0.50

Course ID: 006880

Calculus 1 for Honours Mathematics

Absolute values and inequalities. Sequences and their limits. Introduction to series. Limits of functions and continuity. The Intermediate Value theorem and approximate solutions to equations. Derivatives, linear approximation, and Newton's method. The Mean Value theorem and error bounds. Applications of the Mean Value theorem, Taylor polynomials and Taylor's theorem, Big-O notation. Suitable topics are illustrated using computer software. [Offered: F,W,S; online: F,W,S]

Prereq: 4U Calculus and Vectors.

Antireq: MATH 116, 117, 127, 147

Also offered Online

MATH 138 LEC,TST,TUT 0.50

Course ID: 006881

Calculus 2 for Honours Mathematics

Introduction to the Riemann integral and approximations. Antiderivatives and the fundamental theorem of calculus. Change of variables, methods of integration. Applications of the integral. Improper integrals. Linear and separable differential equations and applications. Tests for convergence for series. Binomial series, functions defined as power series and Taylor series. Vector (parametric) curves in R2. Suitable topics are illustrated using computer software. [Offered: F,W,S; online: F,W,S]

Prereq: (MATH 116 or 117 or 127 with a grade of at least 70%) or MATH 137 with a grade of at least 60% or MATH 147.

Antireq: MATH 118, 119, 128, 148

Also offered Online

MATH 145 LEC,TST,TUT 0.50

Course ID: 006886

Algebra (Advanced Level)

MATH 145 is an advanced-level version of MATH 135. [Offered: F]

Department Consent Required

Prereq: 4U Calculus and Vectors or 4U Mathematics of Data Management; Honours Mathematics students only.

Antireq: MATH 135

MATH 146 LAB,LEC,TST,TUT 0.50

Course ID: 006887

Linear Algebra 1 (Advanced Level)

MATH 146 is an advanced-level version of MATH 136.

[Note: Students who receive a minimum grade of 90% in MATH 135 may contact the instructor of MATH 146 to seek admission without the formal prerequisites. Offered: W]

Prereq: MATH 145; Honours Mathematics students only.

Antireq: MATH 106, 114, 115, 136, NE 112

MATH 147 LAB,LEC,TST,TUT 0.50

Course ID: 006888

Calculus 1 (Advanced Level)

MATH 147 is an advanced-level version of MATH 137. [Offered: F]

Department Consent Required

Prereq: 4U Calculus and Vectors; Honours Mathematics students only.

Antireq: MATH 116, 117, 124, 127, 137

MATH 148 LEC,TST,TUT 0.50

Course ID: 006889

Calculus 2 (Advanced Level)

MATH 148 is an advanced-level version of MATH 138.

[Note: Students who receive a minimum grade of 90% in MATH 137 may contact the instructor of MATH 148 to seek admission without the formal prerequisites. Offered: W]

Prereq: MATH 147; Honours Mathematics students only.

Antireq: MATH 118, 119, 128, 138

MATH 199 LEC 0.50

Course ID: 015595

Mathematical Discovery and Invention

A course in problem solving in which intriguing and difficult problems are solved. Problems are taken mainly from the elementary parts of applied mathematics, computer science, statistics and actuarial science, pure mathematics, and combinatorics and optimization. Material relevant to the problems is taught in depth.

Instructor Consent Required

Only offered Online

MATH 200s

MATH 207 LEC,TST,TUT 0.50

Course ID: 013105

Calculus 3 (Non-Specialist Level)

Multivariable functions and partial derivatives. Gradients. Optimization including Lagrange multipliers. Polar coordinates. Multiple integrals. Surface integrals on spheres and cylinders. Introduction to Fourier Series. [Offered: F,W,S]

Prereq: MATH 128 or 138 or 148.

Antireq: AMATH 231, MATH 212, 212N/NE 217, MATH 217, 227, 237, 247

Also offered Online

MATH 211 LEC,TST,TUT 0.50

Course ID: 006891

Advanced Calculus 1 for Electrical and Computer Engineers

Fourier series. Ordinary differential equations. Laplace transform. Applications to linear electrical systems. [Offered: F,W]

Prereq: MATH 119; Not open to Mathematics students.

Antireq: AMATH 350, MATH 218, 228

(Cross-listed with ECE 205)

MATH 212 LEC,TST,TUT 0.50

Course ID: 006892

Adv Calculus 2 for Electrical Engineers

Triple integrals, cylindrical and spherical polar coordinates. Divergence and curl, applications. Surface integrals, Green's, Gauss' and Stokes' theorems, applications. Complex functions, analytic functions, contour integrals, Cauchy's integral formula, Laurent series, residues. [Offered: F,S]

Prereq: MATH 211/ECE 205; Not open to Mathematics students.

Antireq: AMATH 231, MATH 207, 217, 227, 237, 247

(Cross-listed with ECE 206)

MATH 213 LEC,TUT 0.50

Course ID: 011849

Signals, Systems, and Differential Equations

Laplace transform methods for: solving linear ordinary differential equations, classical signals, and systems. Transfer functions, poles, and zeros; system stability. Frequency response of linear systems and its log-scale representation (Bode plot). Fourier series. Applications in areas of interest for software engineers and computer scientists. Brief introduction to Fourier transforms in the context of signals and systems.

Prereq: One of MATH 118, MATH 119, MATH 128, MATH 138.

Antireq: AMATH 250, 251, MATH 211/ECE 205, 218, 228

MATH 215 LEC,TUT 0.50

Course ID: 013464

Linear Algebra for Engineering

Systems of linear equations; their representation with matrices and vectors; their generalization to linear transformations on abstract vector spaces; and the description of these linear transformations through quantitative characteristics such as the determinant, the characteristic polynomial, eigenvalues and eigenvectors, the rank, and singular values. [Offered: F,W]

Prereq: Level at least 2A Computer Engineering or Electrical Engineering students only.

Antireq: MATH 106, 114, 115, 136, 146, NE 112

MATH 217 LEC,TUT 0.50

Course ID: 006897

Calculus 3 for Chemical Engineering

Curves and surfaces in R3. Multivariable functions, partial derivatives, the chain rule, gradients. Optimization, Lagrange Multipliers. Double and triple integrals, change of variable. Vector fields, divergence and curl. Vector integral calculus: Green's theorem, the Divergence theorem and Stokes' theorem. Applications in engineering are emphasized. [Offered: F,W]

Prereq: MATH 118; Not open to Mathematics students.

Antireq: AMATH 231, CIVE 221, ENVE 221, MATH 207, 212/ECE 206, 227, 237, 247, MATH 212N/NE 217, ME 201

MATH 218 LEC,TUT 0.50

Course ID: 006898

Differential Equations for Engineers

First order equations, second order linear equations with constant coefficients, series solutions, the Laplace transform method, systems of linear differential equations. Applications in engineering are emphasized. [Offered: F,S]

Prereq: One of MATH 118, 119, 128, 138, 148, SYDE 112; Engineering or Earth Science students only.

Antireq: AMATH 250, 251, 350, 351, CIVE 222, ENVE 223, MATH 211/ECE 205, 228, MATH 212N/NE 217, ME 203, SYDE 211

MATH 225 LEC,TUT 0.50

Course ID: 006870

Applied Linear Algebra 2

Vector spaces. Linear transformations and matrices. Inner products. Eigenvalues and eigenvectors. Diagonalization. Applications. [Offered: F,S; online: W]

Prereq: MATH 106 or 136 or 146.

Antireq: MATH 235, 245

Also offered Online

MATH 227 LEC,TUT 0.50

Course ID: 006907

Calculus 3 for Honours Physics

Directional derivative and the chain rule for multivariable functions. Optimization, Lagrange multipliers. Double and triple integrals on simple domains; transformations and Jacobians; change of variable in multiple integrals. Vector fields, divergence and curl. Vector integral calculus: Line and surface integrals, Green's Theorem, Stokes' Theorem, Gauss' Theorem, conservative vector fields. [Offered: F]

Prereq: MATH 128 or 138; Only open to Science students in honours plans.

Antireq: AMATH 231, MATH 207, 212/ECE 206, 217, 237, 247, MATH 212N/NE 217

Also offered Online

MATH 228 LEC,TUT 0.50

Course ID: 006908

Differential Equations for Physics and Chemistry

First-order equations, second-order linear equations with constant coefficients, series solutions and special functions, the Laplace transform method. Applications in physics and chemistry are emphasized. [Offered: F,W; online: W,S]

Prereq: MATH 128 or 138; Not open to Mathematics students.

Antireq: AMATH 250, 251, 350

Also offered Online

MATH 229 LEC,TUT 0.50

Course ID: 013104

Introduction to Combinatorics (Non-Specialist Level)

Introduction to graph theory: colourings, connectivity, Eulerian tours, planarity. Introduction to combinatorial analysis: elementary counting, generating series, binary strings. [Offered: F]

Prereq: (MATH 106 or 136 or 146) and (MATH 135 or 145).

Antireq: CO 220, MATH 239, 249

MATH 235 LEC,TST,TUT 0.50

Course ID: 006913

Linear Algebra 2 for Honours Mathematics

Orthogonal and unitary matrices and transformations. Orthogonal projections, Gram-Schmidt procedure, best approximations, least-squares. Inner products, angles and orthogonality, orthogonal diagonalization, singular value decomposition, applications. [Offered: F,W,S]

Prereq: (MATH 106 or 114 or 115 with a grade of at least 70%) or (MATH 136 with a grade of at least 60%) or MATH 146; Honours Mathematics or Mathematical Physics students.

Coreq: MATH 128 or 138 or 148.

Antireq: MATH 225, 245

Also offered Online

MATH 237 LEC,TST,TUT 0.50

Course ID: 006914

Calculus 3 for Honours Mathematics

Calculus of functions of several variables. Limits, continuity, differentiability, the chain rule. The gradient vector and the directional derivative. Taylor's formula. Optimization problems. Mappings and the Jacobian. Multiple integrals in various co-ordinate systems.

[Note: MATH 247 may be substituted for MATH 237 whenever the latter is a plan requirement. Offered: F,W,S]

Prereq: (One of MATH 106, 114, 115, 136, 146) and (MATH 128 with at least 70% or MATH 138 with at least 60% or MATH 148); Honours Math or Math/Physics students.

Antireq: MATH 207, 212/ECE 206, MATH 212N, 217, 227, 247

Also offered Online

MATH 239 LEC,TST,TUT 0.50

Course ID: 006915

Introduction to Combinatorics

Introduction to graph theory: colourings, matchings, connectivity, planarity. Introduction to combinatorial analysis: generating series, recurrence relations, binary strings, plane trees. [Offered: F,W,S]

Prereq: ((MATH 106 with a grade of at least 70% or MATH 136 or 146) and (MATH 135 with a grade of at least 60% or MATH 145)) or level at least 2A Software Engineering; Honours Mathematics students only.

Antireq: CO 220, MATH 229, 249

Also offered Online

MATH 245 LEC,TST 0.50

Course ID: 006920

Linear Algebra 2 (Advanced Level)

MATH 245 is an advanced-level version of MATH 235. [Offered: F,S]

Prereq: MATH 146; Honours Mathematics students only.

Antireq: MATH 225, 235

MATH 247 LEC,TST 0.50

Course ID: 006921

Calculus 3 (Advanced Level)

Topology of real n-dimensional space: completeness, closed and open sets, connectivity, compact sets, continuity, uniform continuity. Differential calculus on multivariable functions: partial differentiability, differentiability, chain rule, Taylor polynomials, extreme value problems. Riemann integration: Jordan content, integrability criteria, Fubini's theorem, change of variables. Local properties of continuously differentiable functions: open mapping theorem, inverse function theorem, implicit function theorem. [Offered: F,W,S]

Prereq: MATH 146, 148; Honours Mathematics students only.

Antireq: MATH 237

MATH 249 LEC,TST 0.50

Course ID: 006922

Introduction to Combinatorics (Advanced Level)

MATH 249 is an advanced-level version of MATH 239. [Offered: F,W]

Prereq: (MATH 135 with minimum grade of 80% or MATH 145) and (MATH 136 or 146); Honours Mathematics students only.

Antireq: CO 220, MATH 229, 239